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- 13413760131
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$$\varepsilon + \cdot =$$

ε

- $\gg$
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$$\binom{3}{}$$

- $\rightarrow$
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-

$$\| - \|_2^2 + \alpha \left[\rho \| \|_1 + \frac{1-\rho}{2} \| \|_2^2 \right]$$

- $\rho \| \|_1 \rightarrow$
- $\frac{1-\rho}{2} \| \|_2^2 \rightarrow$
- $\rho \rightarrow 1 \rho \rightarrow 0$

- $\lambda_2 > 0 \rightarrow$
-
-

-
-
-
-

- $\rightarrow$
- $\rightarrow$
- $\rightarrow$

- 2
-
- α

- `ElasticNet(positive=True)`
- $\alpha \in [10^{-7}, 10^{0}] \rho \in [0.01, 0.99]$
-
- $|\beta_0/\hat{}| > 10$

- $= \oslash$
-

α

- $= 100\alpha \in [10^{-6}, 10^{0}]$
- $= 500\alpha \in [10^{-5}, 10^{1}]$
- $= 1000\alpha \in [10^{-4}, 10^{2}]$
- `l1_ratio`$\approx 0.99 \rightarrow$

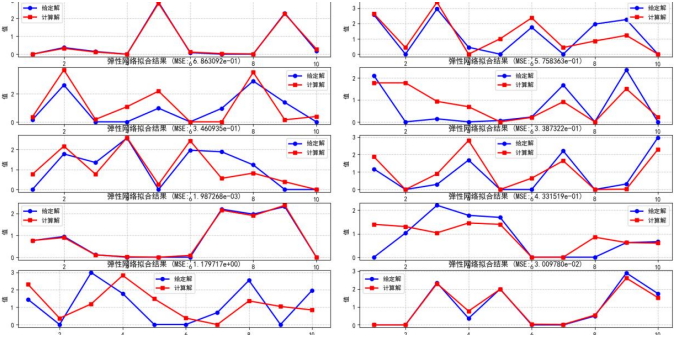
- `positive=True`
-
-

- $|\beta_0/\hat{}| < 10$
- $|\beta_0/\hat{}| \gg 10$
-

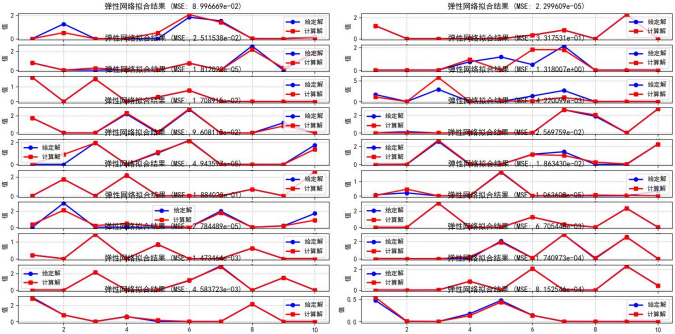
- 2
-
- α

- `ElasticNet(positive=True)`
- $\alpha \in [10^{-7}, 10^{0}] \rho \in [0.01, 0.99]$
-
- $|\beta_0/\hat{}| > 10$

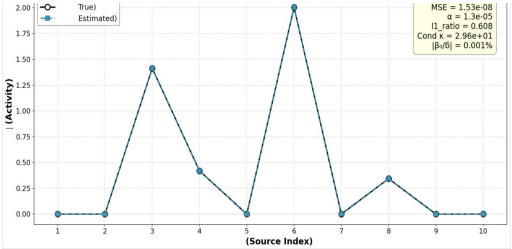
- $= \oslash$
-

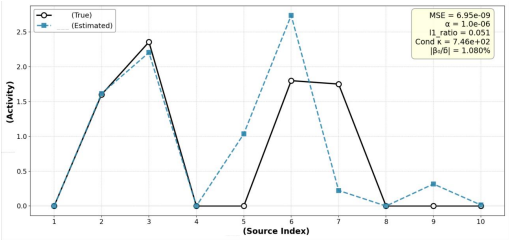

弹性障碍拟合结果 (MSE: 6.863092e-01)
弹性障碍拟合结果 (MSE: 5.758363e-01)
弹性障碍拟合结果 (MSE: 3.460935e-01)
弹性障碍拟合结果 (MSE: 3.387322e-01)
弹性障碍拟合结果 (MSE: 1.987268e-03)
弹性障碍拟合结果 (MSE: 4.331519e-01)
弹性障碍拟合结果 (MSE: 1.179717e+00)
弹性障碍拟合结果 (MSE: 3.009780e-02)
给定解
计算解

$10^{-4}10^{-9}$


弹性网络拟合结果 (MSE: 8.996669e-02)
弹性网络拟合结果 (MSE: 2.511538e-02)
弹性网络拟合结果 (MSE: 1.812908e-05)
弹性网络拟合结果 (MSE: 1.708715e-02)
弹性网络拟合结果 (MSE: 9.608115e-02)
弹性网络拟合结果 (MSE: 4.943078e-06)
弹性网络拟合结果 (MSE: 1.884628e-01)
弹性网络拟合结果 (MSE: 7.784489e-05)
弹性网络拟合结果 (MSE: 1.473466e-03)
弹性网络拟合结果 (MSE: 4.583723e-03)
弹性网络拟合结果 (MSE: 2.299609e-05)
弹性网络拟合结果 (MSE: 3.317531e-01)
弹性网络拟合结果 (MSE: 1.318007e+00)
弹性网络拟合结果 (MSE: 4.220029e-35)
弹性网络拟合结果 (MSE: 2.569759e-02)
弹性网络拟合结果 (MSE: 1.863430e-02)
弹性网络拟合结果 (MSE: 1.063408e-05)
弹性网络拟合结果 (MSE: 6.705408e-03)
弹性网络拟合结果 (MSE: 1.740973e-04)
弹性网络拟合结果 (MSE: 8.152046e-04)
拟定解
计算解
值

10^{-9}

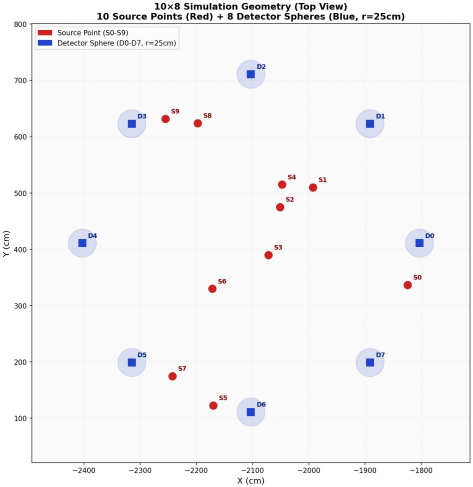

True)
Estimated)
MSE = 1.53e-08
α = 1.3e-05
l1_ratio = 0.608
Cond κ = 2.96e+01
|βₛ/β| = 0.001%
(Activity)
(Source Index)


(True)
(Estimated)
MSE = 6.95e-09
α = 1.0e-06
l1_ratio = 0.051
Cond κ = 7.46e+02
|βₛ/β| = 1.080%
(Activity)
(Source Index)

- $10^8 \times$
- γ

$$= 20$$
$$4 \times 3 = 12$$

$$8 \times 12$$

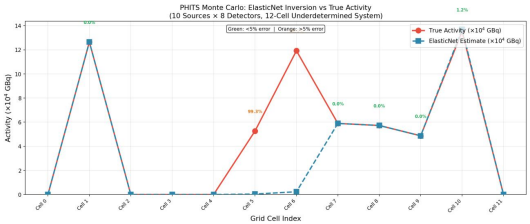


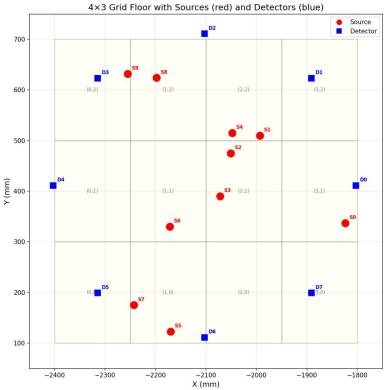
-
- $4 \times 3 = 12$
-
-
- 8×12

$\alpha = 1.0 \times 10^{-6}$ `l1_ratio`$= 0.99^2 = 0.463$

1.265×10^5	1.265×10^5
5.266×10^4	3.93×10^2
1.192×10^5	2.31×10^3
5.894×10^4	5.894×10^4
5.724×10^4	5.724×10^4
4.870×10^4	4.870×10^4
1.352×10^5	1.373×10^5

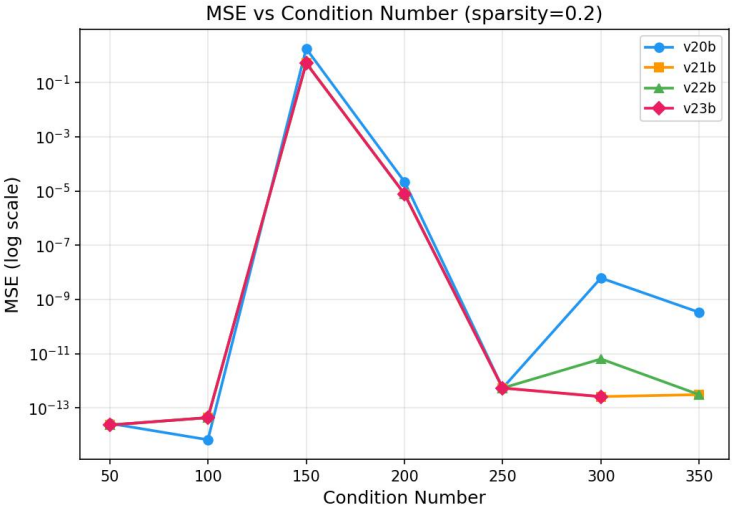
< 2

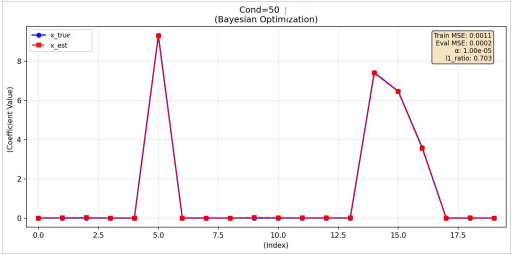

PHITS Monte Carlo: ElasticNet Inversion vs True Activity
(10 Sources × 8 Detectors, 12-Cell Underdetermined System)
Green: <5% error | Orange: >5% error
True Activity (×10⁶ GBq)
ElasticNet Estimate (×10⁶ GBq)
Activity (×10⁶ GBq)
Grid Cell Index
Cell 0
Cell 1
Cell 2
Cell 3
Cell 4
Cell 5
Cell 6
Cell 7
Cell 8
Cell 9
Cell 10
Cell 11
0.0%
99.3%
5.8%
5.6%
4.8%
1.2%

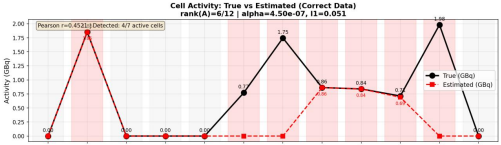

4×3 Grid Floor with Sources (red) and Detectors (blue)
Source
Detector
Y (mm)
X (mm)
D0
D1
D2
D3
D4
D5
D6
D7
S0
S1
S2
S3
S4
S5
S6
S7
S8
S9
(0,2)
(1,2)
(2,2)
(3,2)
(0,1)
(1,1)
(2,1)
(3,1)
(0,0)
(1,0)
(2,0)
(3,0)

- 10^{-9}
- 10^{-10}
-

> -
> -
> -


MSE vs Condition Number (sparsity=0.2)
v20b
v21b
v22b
v23b
MSE (log scale)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
50
100
150
200
250
300
350
Condition Number


Cond=50 |
(Bayesian Optimization)
x_true
x_est
Train MSE: 0.0011
Eval MSE: 0.0002
α: 1.00e-05
l1_ratio: 0.703
(Coefficient Value)
(Index)


Cell Activity: True vs Estimated (Correct Data)
rank(A)=6/12 | alpha=4.50e-07, l1=0.051
Pearson r=0.4521 | Detected: 4/7 active cells
Activity (GBq)
True (GBq)
Estimated (GBq)

$\approx$

-
-
- `l1_ratio`≈ 0.99

$$\frac{\qquad\qquad}{\qquad\qquad}$$

$$510\times$$

$$\frac{\qquad\qquad}{\qquad\qquad}$$

α

< 2

-
- $\rightarrow$

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-
-

-
-
-